User Guide: Integrated Interpolation Suite (GUI)

Group No: 3

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1 Introduction

Welcome to the Integrated Interpolation Suite! This application replaces complex manual calculations with a simple, easy-to-use Windows interface. It allows you to estimate unknown values from experimental data using five advanced mathematical methods.

Unlike the previous version, this Phase 3 release features a full Graphical User Interface (GUI), meaning you no longer need to type commands. You can simply point, click, and view your results instantly.

2 System Requirements

To install and run this application, your computer needs:

- **Operating System:** Windows XP, Windows 7, or newer.
- **Prerequisites:**
  - Microsoft .NET Framework 4.0.
  - Microsoft Visual C++ 2010 Redistributable Package.

(Note: The installer will attempt to detect and install these automatically if they are missing.)

3 Installation Guide

The software comes with a professional installer. Follow these steps to set it up:

Step 1: Locate the Installer
Navigate to the distribution folder provided by Group 3. You should see two files:

- `setup.exe` (The Bootstrapper)
- `InterpolationSuiteInstaller.msi` (The Main Installer)

Step 2: Run the Setup
Double-click on **`setup.exe`**. This ensures all system requirements are checked before installation begins.

Step 3: Follow the Wizard
An installation wizard will appear. Click "Next" through the prompts. You can choose the default installation folder (recommended).

Step 4: Launch the Application
Once finished, you can start the program from:

- The shortcut on your **Desktop**.
- The **Start Menu** under "CS346 Group 3".

4 How to Use the Application

The interface is designed to be intuitive. Follow this workflow to perform a calculation.

Step 1: Define the Data Size
At the top of the window, locate the text box labeled **"N"**. Enter the number of data points you have (e.g., `3`). The data grid will automatically resize to provide the correct number of rows.

Step 2: Enter Your Data
Type your known data into the grid:

- **Column X:** Enter your independent variables (e.g., Time).
- **Column Y:** Enter your dependent variables (e.g., Distance).

Note: The system will automatically sort your data by X-value to ensure accurate calculations.

Step 3: Select a Method
Use the dropdown menu labeled **"Interpolation Method"** to choose one of the five algorithms (e.g., *Linear Interpolation*).

Step 4: Enter the Target
In the **"Target X"** box, type the X-value for which you want to find the estimated Y-value.

Step 5: Calculate
Click the **"Calculate"** button. The result will appear instantly in the result box at the bottom.

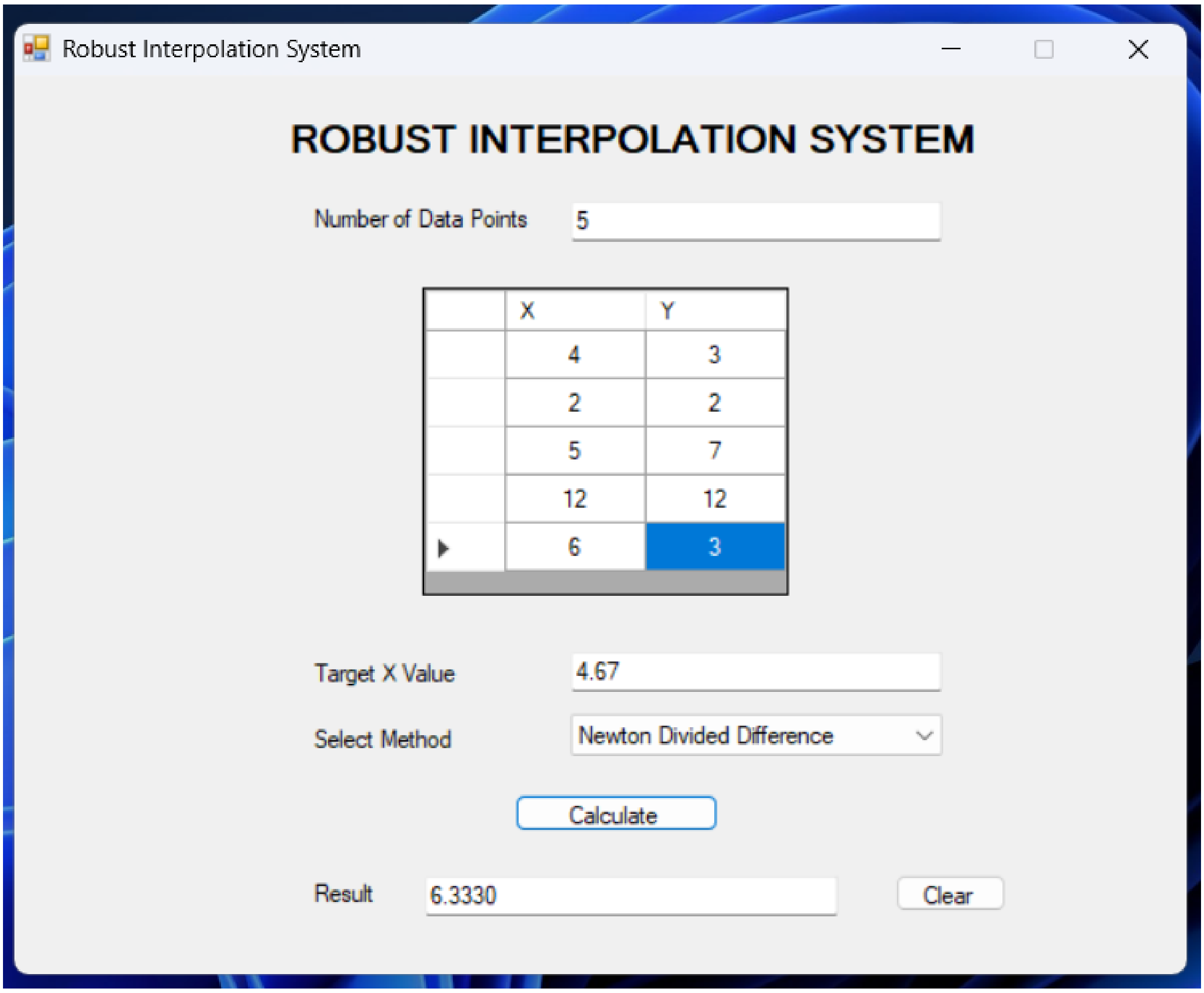

Figure 1: A screenshot of the application GUI

5 Understanding the Methods

The application offers five ways to analyze your data:

- **Piecewise Constant:** Assumes the value stays the same until the next data point. Useful for step-like data (e.g., tax brackets).
- **Nearest Neighbor:** Simply picks the value of the closest known point. Best for rough estimates where smoothness doesn't matter.
- **Linear Interpolation:** Draws a straight line between the two nearest points. This is the standard "connect-the-dots" method.
- **Newton's Divided Difference:** Creates a smooth polynomial curve that passes through every single point. Great for scientific data.
- **Lagrange Polynomial:** Mathematically equivalent to Newton's method but calculated differently. It also produces a smooth curve.

6 Troubleshooting Errors

The application includes robust error handling. Here are common messages you might see:

Message: `"Please fill all X and Y values."` You left a cell empty in the grid. All rows must contain numbers.

Message: `"Invalid X value at row..."` You typed text (like ”abc”) instead of a number. Please check your inputs.

Message: `"Duplicate X values detected!"` You entered the same X-value twice (e.g., Time=5.0 appears in two rows). This is mathematically impossible for functions. Please remove the duplicate.

Message: `"Target X is outside the data range..."` You are trying to guess a value outside your known data (Extrapolation).

- **Linear/Piecewise:** Will stop and show an Error (cannot extrapolate).
- **Newton/Lagrange:** Will show a Warning but give you a result (use with caution).

7 Uninstallation

To remove the software:

Step 1: Go to the Windows **Control Panel**.

Step 2: Select **”Uninstall a program”**.

Step 3: Find **”Interpolation Suite”** in the list.

Step 4: Right-click and select **Uninstall**.